RAJALAKSHMI ENGINEERING COLLEGE

Monish A 251001155-2025-IT

M2

Started on	Tuesday, 4 August 2026, 8:29 AM
State	Finished
Completed on	Tuesday, 11 August 2026, 8:14 AM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int function(int n){
int i=1;
int s=1;
int count=0;
while(s<=n){
    count++;
i++;
count++;
s+=i;
count++;
}
return count;
}
int main(){
int n;
scanf("%d",&n);
int res=function(n);
res=res+3;
printf("%d",res);
}
// #include <stdio.h>

// int main()
// {
// int n;
// scanf("%d", &n);
// int i = 1;
// int s = 1;
// int count = 0;
// while (1)
```

```
// {count++;
// if (!(s <= n))
// break;
// i++;
// count++;
// s += i;
// count++;
//  }
// printf("%d", count);
//  return 0;
// }
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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